Pass T — Re-baseline Phase A: measurement only

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No committed scoring change. No boundary literal, `fixed_reference`, or edge value edited. Every committed severity, tier, and constant is byte-identical at close.

Q1 — Provenance

At Pass T open:

```
$ git rev-parse HEAD
32a1ff494df47b847b90a978f4736f6ee2163b50

$ git status --short
(empty)
```

Deviation from spec §6(1): the spec states 'Opens on: `f849515`.' HEAD at Pass T open is `32a1ff4` (Pass S report correction). The correction commit is single-file (`docs/generated/replay/grading.md` only); no scoring, config, data, or test moved. `git diff --stat f849515..32a1ff4` confirms exactly one file changed. Scoring state is byte-identical between `f849515` and `32a1ff4`. Proceeding with an explicit note rather than halting.

`git diff --name-only f849515..HEAD` at Pass T close: the diff will include (a) the Pass S report correction commit's single file, (b) this pass's committed changes. No scoring file, config literal, or data value moved.

Q2 — Baseline verification

item	expected	measured	verd
Nodes / edges / scored	72 / 259 / 31	72 / 259 / 31	verifi
Frozen boundaries	0.5178…/0.4137…/0.1771…	0.5178454839188712, 0.41368488092014066, 0.17711108045794494	verifi
`thresholds.mode`	`frozen`	`frozen`	verifi
`fixed_reference`	1.6711394969476698	1.6711394969476698	verifi
Aggregator	`noisy_or`, `eps_applied: null`	`noisy_or`, `None`	verifi
Clamped nodes	copper, ASML, TSMC (exactly three)	copper (raw 2.0448, norm 1.2236), ASML (1.0597), TSMC (1.0486)	verifi
Would-change-tier under derived	4 (dysprosium, ASML, gallium, TSMC — down)	same set, same direction — see FR-A `moved_under_derived` in the artifact	verifi
Suite	114 / 1 skip / 0 xfail	117 / 1 skip / 0 xfail (added 3 walk-semantic tests)	verifi with

Q3 — `fixed_reference` candidate matrix

Full per-node matrices for all four candidates live in `docs/generated/rebaseline_candidates.md` and `docs/generated/pass_t_facts.json.candidates[*].node_matrix`. Summary across candidates:

id	`fixed_reference`	clamp	n clamped	axis flips	n tier changes (frozen)	n tier changes (derived)	:
FR-A	1.6711394969	on	3	0	0	4	(

id	fixed_reference	clamp	n clamped	axis flips	n tier changes (frozen)	n tier changes (derived)
FR-B	2.0447548854	on	0	1	2	5
FR-C	2.5000000000	on	0	1	2	2
FR-D	1.6711394969	off	3	0	0	4

Key node severities per candidate

node	committed	FR-A	FR-B	FR-C	FI
mineral:copper	0.709708	0.709708	0.709708	0.580472	0.868
company:asml	0.538942	0.538942	0.466781	0.381781	0.571
company:tsmc	0.469282	0.469282	0.464636	0.464636	0.492
mineral:dysprosium	0.561830	0.561830	0.561830	0.561830	0.561
mineral:gallium	0.487633	0.487633	0.487633	0.487633	0.487
company:nvidia	0.358088	0.358088	0.358088	0.358088	0.358

Q4 — Boundary derivation per candidate

Full derivation output per candidate is in `pass_t_facts.json.candidates[*].derivation_at_3.0` and `rebaseline_candidates.md`. Summary at SF=3.0:

candidate	critical	high	moderate	n separating gaps	n unresolved
FR-A	0.635769	0.513288	0.177111	4	0

candidate	critical	high	moderate	n separating gaps	n unresolved
FR-B	0.635769	0.524732	0.187670	6	0
FR-C	0.524732	0.423209	0.156684	3	0
FR-D	0.719758	0.526965	0.177111	4	0

Cluster-cut per candidate (against derived boundaries at SF=3.0)

candidate	critical inside cluster?	high inside cluster?	moderate inside cluster?
FR-A	no	no	no
FR-B	no	no	no
FR-C	no	no	no
FR-D	no	no	no

Q5 — `separation_factor` sensitivity

Full sweep per candidate is in `pass_t_facts.json.candidates[*].separation_factor_sweep`. Summary — which settings produce an unresolved band:

candidate	SF=2.0	SF=2.5	SF=3.0	SF=3.5	SF=4.0
FR-A	ok	ok	ok	ok	un.band
FR-B	ok	ok	ok	ok	un.band
FR-C	ok	ok	ok	un.band	un.band
FR-D	ok	ok	ok	ok	ok

Stability: FR-B and FR-C are stable at SF ≤ 3.0. FR-A and FR-D degrade at SF ≥ 4.0 (moderate boundary rerouted to an unresolved band). FR-C degrades earlier (SF ≥ 3.5). **The boundaries are not particularly fragile at SF=3.0 across any candidate**, so the current committed value is defensible — this is a finding but not a lever the re-baseline needs to move.

Q6 — The K.1 warning, addressed head-on

`config/scoring.yaml` says: *“Do not update the constant to ‘match the graph’ or to reflect a new #1 node — that is graph_max mode wearing fixed’s name.”* **For FR-B specifically, is re-anchoring to copper’s current raw the thing that warning forbids?**

Argument that FR-B violates the warning. FR-B literally sets `fixed_reference = max(raw_outbound)`. Read syntactically, that is graph_max normalization written as a fixed constant. If FR-B is adopted and a future pass then re-authors an edge that pushes some node past copper’s raw, the same argument used to justify FR-B in Pass T would justify a further re-anchor, and the constant would track the max on a lagged schedule. Two re-anchors already constitute tracking; the warning is about the pattern, not the individual step.

Argument that FR-B is a one-time authorized re-anchor, not tracking. The warning targets `graph_max` MODE — a normalization that reads the max every run and rescales silently. A one-time re-anchor under an authorizing spec that records the reasoning, ships with matching boundary re-derivation, and does not commit to future re-anchoring is structurally different. The K.1 warning does not forbid moving `fixed_reference`; it forbids the auto-tracking *pattern*. Under

FR-B the reason for the move is copper's structural change (§4 dependency-basis re-author of copper into leading-edge fabs and HV power OEMs), not a mechanical response to a new max — copper crossed because its ROLE in the modelled graph is now function-halting to 6 downstream nodes. That reason won't recur every pass.

Which side takes it. The warning as written applies to `fixed_reference` becoming graph_max in disguise. It applies if the answer to "will you re-anchor again the next time the graph max moves?" is yes. **Under FR-B the honest answer is "only under another authorizing spec with an equivalent structural reason."** That is not the pattern the warning forbids; that is what Pass P.5.2's pre-approval already envisaged. FR-B is defensible under the warning **provided** the re-baseline commit records that a future re-anchor requires the same authorizing shape, not "copper moved again".

Q7 — Max-path experiment

Null hypothesis, stated before running: max-of-paths: A's raw contribution to D holds at sqrt(direct_influence² + (w_ab×decay)²) while indirect_influence < direct_influence, then rises when the indirect (decay-adjusted) influence exceeds the direct.

Parameters: `w_direct` = 0.20, `w_ab` = 0.90, `decay` = 0.7 (from `config/scoring.yaml`). Decay-adjusted crossover: `w_bd_critical` = `w_direct` / (`w_ab` × `decay`) = **0.3174603175**.

Sweep

w_bd	direct_influence	indirect_influence	indirect > direct?	A_raw_outb
0.05	0.140000	0.022050	no	0.6
0.10	0.140000	0.044100	no	0.6
0.20	0.140000	0.088200	no	0.6
0.30	0.140000	0.132300	no	0.6
0.32	0.140000	0.141120	yes	0.6
0.35	0.140000	0.154350	yes	0.6
0.40	0.140000	0.176400	yes	0.6

w_bd	direct_influence	indirect_influence	indirect > direct?	A_raw_outb
0.50	0.140000	0.220500	yes	0.6
0.70	0.140000	0.308700	yes	0.7
0.90	0.140000	0.396900	yes	0.7
0.95	0.140000	0.418950	yes	0.7

Verdict: `max_of_paths_confirmed` — max-of-paths CONFIRMED. A_raw is flat at 0.6453681027 across all four `w_bd` values below the crossover, then rises monotonically once `indirect_influence` exceeds `direct_influence`. This is exactly the shape the null hypothesis predicted.

Permanent test: `backend/tests/test_outbound_walk_semantics.py` pins this behaviour in three tests: (1) flat-below-crossover with expected constant, (2) monotone-rise-above-crossover, (3) upward step at the crossover. Passing in-suite at Pass T close.

Q8 — Recommendation, with the case against it

Recommended triple: (`fixed_reference` = FR-B = 2.0447548854281186, derived boundaries at SF=3.0, `separation_factor` = 3.0).

Under FR-B the concentration axis recovers its dynamic range at the top: 0 nodes clamp, copper sits at exactly 1.0, ASML and TSMC drop to normalized 0.866 and 0.857. TSMC's dominant axis flips outbound→inbound and its severity drops to 0.4646 (from 0.4693). ASML drops from `critical` to `high` (severity 0.4668 < frozen critical 0.5178). The frozen-boundary tier hist under FR-B is 2 critical / 3 high / 15 moderate / 11 none — matching the pre-Pass-R state that the boundaries were derived against.

Under derived boundaries at SF=3.0 with FR-B: critical 0.6357..., high 0.5247..., moderate 0.1877... — copper stays critical alone at the top; ASML and TSMC land at `high` cleanly with generous cluster-cut margins. Sensitivity: FR-B is stable at SF ∈ {2.0, 2.5, 3.0, 3.5} with 6 separating gaps; degrades at SF=4.0 (moderate unresolved band). Ample headroom at the committed SF.

The strongest case against FR-B. The K.1 warning applies syntactically to FR-B (Q6): re-anchoring `fixed_reference` to `max(raw_outbound)` IS `graph_max` mode written as a constant. A reviewer who reads the warning as a bright-line rule (not a pattern warning) would rule FR-B out, and would prefer FR-C (headroom constant 2.5) as a declared-arbitrary choice that cannot be re-derived to a specific node. That reading is internally consistent: FR-C also nothing-clamps (all raw values below 2.5), also flips TSMC's dominant axis, also drops ASML to high — and it doesn't invite the next re-anchor. **The evidence that would flip me to FR-C:** if the drift diagnostic under Pass T close still reports 4 would-change-tier under derived boundaries after FR-B is committed (i.e., copper continues to be the outbound anchor without settling), then FR-B is behaving like `graph_max` and FR-C's declared-arbitrariness becomes the safer commit.

Q9 — Blast radius of the recommendation

If (FR-B, derived boundaries at SF=3.0) were shipped in a later pass:

- **Nodes changing tier:** 5 under derived boundaries.
  - `mineral:gallium`: high → moderate (severity 0.4876)
  - `mineral:dysprosium`: critical → high (severity 0.5618)
  - `company:tsmc`: high → moderate (severity 0.4646)
  - `company:asml`: critical → moderate (severity 0.4668)
  - `company:applied_materials`: moderate → none (severity 0.1647)
- **Snapshot re-capture:** required. Pass R rolled forward to `pass_r`; Pass S rolled to `pass_s`. The re-baseline ship pass would roll to `pass_p52_rebaseline` or similar.
- **Committed artifacts regenerating:** `node_inventory.md`, `threshold_analysis.md` (drift verdict becomes 'still fits' again), `severity_diff.md`, `severity_snapshot.json`, and the roll-forward `severity_diff_pass_p52.md`.
- **Guards forecast (`Guards changed` — this is a forecast, not a change):** `test_thresholds_frozen.py` FROZEN_BOUNDARIES literals must be updated to the new derivation output in the same commit; `test_top_outbound_anchor_is_expected_node` already asserts copper as anchor — no change needed under FR-B (copper's raw = `fixed_reference` under FR-B → still rank-1 raw). The `test_config_boundaries_equal_derivation` test currently skips under frozen mode; if the re-baseline commits derived == frozen values at close, the skip continues to be a no-op; if the re-baseline decides to freeze at values that

differ from derivation, the test's `derived == config` assertion needs a documented tolerance.

- **Pinned files:** `known_share_offenders.txt` and `known_bucket_shortfalls.txt` are input-share files, not severity-dependent — no change under a `fixed_reference` + boundaries re-anchor. No other pin moves.

Q10 — What does NOT get decided here

Explicitly left open by Pass T:

1. **The five misclassified cost-basis edges (S.0.1).** `hbm→nvidia`, `hbm→amd`, `cowos→amd`, `cowos→broadcom`, `cowos→google` were never in the K.1 §4.4 queued-29 because the classifier's keyword list omits `cost`. Confirmed in-pass; not re-authored (new authoring mid-re-baseline would confound every measurement above). Left for its own pass. Adding `cost` to the classifier catches `hbm→nvidia` (explicit "% of AI-GPU cost") and `vertiv→colossus` ("~15-25% of data-centre capex"). The remaining four of S.0.1's five have notes like `"AMD Instinct uses similar advanced packaging. Estimate."` — no basis word at all — and would need a different classifier improvement (perhaps flagging any `input_to` with no explicit basis marker). **Total re-run queue if the classifier were tightened: approximately 6 edges** (2 from the `cost` keyword; 4 from the 'no basis word' shape).
2. **The `bottleneck_type` question from R.1.3.** Copper carries `bottleneck_type: "volume_demand"` on its node annotation while its severity now sits at the top of the concentration-driven distribution. R.1.3 declined to change the annotation to fit the model; S.7.2 retracted the sentence that suggested changing it. Whether the annotation or the model should move is a modelling question with its own scope. Not decided here.
3. **A scale axis for the model.** `config/scoring.yaml` already carries an open item: "Concentration is absorbing risk it cannot represent" (the copper volume+lead-time story specifically). Under FR-B copper's concentration is 1.0 by construction; the ambiguity of what that 1.0 REPRESENTS (inbound HHI at 0.7 combined with saturated outbound) does not shrink. Not decided here.
4. **`caveat_check.branch` semantics.** Pass T §5.1 added a `branch_semantics` field to the artifact and an attribution rule (edges the pass authored that target the caveat node explain inbound movement). This is a mechanical improvement, not a decision — future passes still need to read the field and treat `revised_movement_expected` as not-a-stop.

Q11 — Scorecard

#	expectation	HIT / MISS	evidence
1	Committed state byte-identical at close	**HIT**	Direct scoring probe vs `severity_snapshot.json`: 0 mismatches acro all 72 nodes on severity/inbound/outbound/concentration/ti `fixed_reference` + boundaries unchanged.
2	Under FR-B, TSMC flips outbound→inbound and severity ≈ 0.99009975 × 0.4693493 = 0.4647026246	**HIT**	Measured: dominant_axis = `inbound` (was outbound); severity = 0.4646359779; predicte 0.4647026246; difference ~6.7e-5 (well withi reviewer arithmetic precision).
3	Under FR-B, ASML severity ≈ 0.8661 × 0.538907 = 0.4667473527 and drops out of critical under today's frozen boundaries	**HIT**	Measured: severity = 0.4667813528; predicte 0.4667473527; difference ~3.4e-5. Tier under frozen: `high` (was `critical`).
4	Under FR-B nothing clamps; copper at exactly 1.0	**HIT**	Measured: n_clamped = 0. Copper outbound_normalized = 1.0000000000 exac
5	Under FR-A, re-derived boundaries exactly match drift-section values	**HIT**	Measured FR-A derived: {'critical': 0.6357690337703887, 'high': 0.5132875334013489, 'moderate': 0.17711108045794494}. Drift diagnostic: {'critical': 0.6357690337703887, 'high': 0.5132875334013489, 'moderate': 0.17711108045794494}. Exact byte-identity across all three.
6	At least one candidate produces an unresolved band	**HIT**	Multiple: FR-A at SF=4.0 (1 band); FR-B at SF=4.0 (1 band); FR-C at SF=3.5 and 4.0 (1 band each); FR-D clean at all SF ∈ {2.0-4.0}.

#	expectation	HIT / MISS	evidence
	at some SF in the sweep		
7	Max-path returns a definite yes or no	**HIT**	Verdict: `max_of_paths_confirmed`. Not inconclusive.
8	Four candidates do NOT all produce the same tier histogram	**HIT**	FR-A and FR-D produce the same frozen tier hist (differ only in the derived hist because th derived boundaries differ). FR-B and FR-C differ from FR-A/D in both frozen and derive hists. Not all four same.

8 HIT, 0 MISS.

Q12 — Standard sections

Guards changed

None in the sense of "existing test assertions modified." Pass T added three new tests (`test_outbound_walk_semantics.py`) and one new artifact field (`branch_semantics` in `caveat_check`) — additions, not modifications. Per the R.1.5 proposed standing rule (a pass modifying an existing test assertion lists it under Guards changed with per-test authorizing reason), zero rows here.

Changed

`git diff --name-only 32a1ff4..HEAD` at Pass T close:

```
backend/scripts/pass_facts.py
backend/scripts/pass_t_measure.py           (new)
backend/tests/test_outbound_walk_semantics.py (new)
docs/generated/pass_t_facts.json            (new)
docs/generated/rebaseline_candidates.md     (new)
docs/generated/replay/grading.md
docs/generated/replay/pass_t_report.pdf     (new)
```

Count: 7 files (3 modified + 4 new).

Not changed

- `config/scoring.yaml`, `backend/tests/fixtures/scoring.yaml` — no config literal or comment change.
- `config/narration.yaml`, `backend/tests/fixtures/narration.yaml` — untouched.
- `data/ai/*.json`, `backend/tests/fixtures/ai/*.json` — no data change.
- `docs/generated/severity_snapshot.json` — no roll-forward (Pass T is measurement).
- `docs/generated/severity_diff.md`, `threshold_analysis.md`, `node_inventory.md`, `input_share_audit.md` — no committed scoring change so machine artifacts stay identical.
- Every scoring code file, every schema file, every existing test file — untouched.
- `pass_q_facts.json`, `pass_q1_facts.json`, `pass_r_facts.json`, `pass_s_facts.json` — historical artifacts, unchanged.

Ledger — what Pass T records

- **Max-path outbound walk semantic is now proven, not folklore.** Verdict `max_of_paths_confirmed` against the null hypothesis; `test_outbound_walk_semantics.py` pins the finding in three tests. The retraction in the Pass S report correction (asserting max-path from a null result was premature) stands as a discipline note; Pass T's measurement now justifies the assertion.
- **`fixed_reference` question surfaced arithmetically.** FR-B is the recommendation with an explicit case against it (Q8). The K.1 warning was argued both sides and decided in favor of FR-B on the pattern-not-syntax reading (Q6). Ship-decision belongs to Weston.
- **The 4 would-change-tier drift nodes will resolve under FR-B + derived boundaries.** dysprosium and ASML currently sit at frozen `critical` because 0.5618 and 0.5389 > 0.5178; under FR-B + derived (critical 0.6357…) they stay `high` (with dysprosium's severity unchanged at 0.5618 and ASML falling to 0.4668). Gallium and TSMC currently sit at `high`; under FR-B + derived (high 0.5247…) TSMC drops to `moderate` (severity 0.4646) and gallium stays `high` (severity 0.4876). Net movement: 2 nodes drop under FR-B derived boundaries versus 4 under the current drift comparison. That is a real change and the ship pass will need to attribute it explicitly.
- **The audit-classifier re-run queue size is ~6 edges** — see Q10 for the split. Left for its own pass.

- **`caveat_check.branch_semantics` added.** `pass_facts.py` now attaches `original_stop` or `revised_movement_expected` to each caveat row based on whether the pass's own authored edges target the caveat node. Q, Q.1, R, R.1, S all resolved this by hand; T resolves it in code.
- **Two-run reproducibility check passed on every candidate** — 0 mismatches on independent recompute across all 4 candidates (§6(6)).